

Local Area Connection Properties



General | Advanced

Internet Protocol (TCP/IP) Properties



General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

- ☐ Obtain an IP address automatically
- ☒ Use the following IP address:

IP address:

10 . 0 . 0 . 10

Subnet mask:

255 . 0 . 0 . 0

Default gateway:

10 . 0 . 0 . 1

- ☐ Obtain DNS server address automatically
- ☒ Use the following DNS server addresses:

Preferred DNS server:

. . .

Alternate DNS server:

. . .

Advanced...

OK

Cancel

Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

15 . 0 . 0 . 10

Subnet mask:

255 . 255 . 0 . 0

Default gateway:

15 . 0 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

. . .

Alternate DNS server:

. . .

Advanced...

OK

Cancel

Internet Protocol (TCP/IP) Properties



General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

162 . 14 . 0 . 10

Subnet mask:

255 . 255 . 0 . 0

Default gateway:

162 . 14 . 0 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

.

Alternate DNS server:

.

Advanced...

OK

Cancel

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#show ip interface brief
*Aug 25 10:33:29.502: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R1#
```

```

R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface ethernet0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Aug 25 10:36:02.372: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#interface ethernet0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#exit
R2#configure terminal
*Aug 25 10:36:35.370: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	20.0.0.2	YES	manual	up	up
Ethernet0/1	30.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R2#

```

```
R3(config-if)#exit
```

```
R3(config)#exit
```

```
R3#
```

```
*Aug 25 11:12:38.641: %SYS-5-CONFIG_I: Configured from console by console
```

```
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	30.0.0.2	YES	manual	up	up
Ethernet0/1	172.19.0.1	YES	manual	up	up
Ethernet0/2	21.0.0.1	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R3#
```

R4#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R4(config)#interface e0/0

R4(config-if)#

*Aug 25 10:43:04.357: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)

R4(config-if)#ip add 15.0.0.1 255.0.0.0

R4(config-if)#no shutdown

R4(config-if)#exit

R4(config)#interface e0/1

R4(config-if)#172.19.0.2 255.255.0.0

% Invalid input detected at '^' marker.

R4(config-if)#ip add 172.19.0.2 255.255.0.0

R4(config-if)#no shutdown

R4(config-if)#exit

R4(config)#exit

R4#

*Aug 25 10:44:45.476: %SYS-5-CONFIG_I: Configured from console by console

R4#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	15.0.0.1	YES	manual	up	up
Ethernet0/1	172.19.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R4#

```
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R5(config)#interface e0/0
R5(config-if)#ip add 21.0.0.2 255.0.0.0
R5(config-if)#no shutdown exit
```

```
% Invalid input detected at '^' marker.
```

```
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#interface e0/1
R5(config-if)#ip add 192.168.168.1 255.255.255.0
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#exit
```

```
R5#configure terminal
```

```
*Aug 25 10:47:18.245: %SYS-5-CONFIG_I: Configured from console by console
```

```
R5#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	21.0.0.2	YES	manual	up	up
Ethernet0/1	192.168.168.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R5#
```

R6#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R6(config)#interface e0/0

R6(config-if)#ip add 162.14.0.1 255.255.0.0

R6(config-if)#no shutdown

R6(config-if)#exit

R6(config)#interface e0/1

R6(config-if)#192.168

*Aug 25 10:49:23.807: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wiz

R6(config-if)#192.168.168.2 255.255.255.0

% Invalid input detected at '^' marker.

R6(config-if)#ip add 192.168.168.2 255.255.255.0

R6(config-if)#no shutdown

R6(config-if)#exit

R6(config)#exit

R6#

*Aug 25 10:50:21.815: %SYS-5-CONFIG_I: Configured from console by console

R6#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	162.14.0.1	YES	manual	up	up
Ethernet0/1	192.168.168.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R6#

```
onfig)#
onfig)#
onfig)#
onfig)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
onfig)#ip route 172.19.0.0 255.255.0.0 20.0.0.2
onfig)#ip route 15.0.0.0 255.0.0.0 20.0.0.2
onfig)#ip route 21.0.0.0 255.0.0.0 20.0.0.2
onfig)#ip route 192.168.168.0 255.255.255.0 20.0.0.2
onfig)#ip route 162.14.0.0 255.255.0.0 20.0.0.2
onfig)#exit
#sh
```

```
Aug 25 10:56:26.772: %SYS-5-CONFIG_I: Configured from console by console
```

```
#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
        20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
```

```
--More--
```

% Invalid input detected at '^' marker.

R2(config)#exit

R2#s

*Aug 25 11:02:04.476: %SYS-5-CONFIG_I: Configured from console by console

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 20.0.0.1
S 15.0.0.0/8 [1/0] via 30.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2

--More--

*Aug 25 11:02:12.328: %IP-4-DUPADDR: Duplicate address 30.0.0.1 on Ethernet0/1, sourced

--More--

```
R4(config)#
R4(config)#
R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
R4(config)#exit
```

```
R4#sh
```

```
*Aug 25 11:07:08.405: %SYS-5-CONFIG_I: Configured from console by console
```

```
R4#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
```

```
--More--
```

```
Ethernet0/2      21.0.0.1      YES manual up
Ethernet0/3      unassigned    YES TFTP   down
```

```
R3#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
```

```
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
```

```
R3(config)#ip route 15.0.0.0 255.0.0.0 172.19.0.2
```

```
R3(config)#ip route 192.168.168.0 255.255.255.0 21.0.0.2
```

```
R3(config)#ip route 162.14.0.0 255.255.0.0 21.0.0.2
```

```
R3(config)#exit
```

```
R3#sh
```

```
*Aug 25 11:16:43.228: %SYS-5-CONFIG_I: Configured from console by console
```

```
R3#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/2
L      21.0.0.1/32 is directly connected, Ethernet0/2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0
L      30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
```

```
-More--
```

Enter configuration commands, one per line. End with CNTL/Z.

R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.1

R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1

R5(config)#ip route 162.14.0.1 255.255.0.0 192.168.168.2

%Inconsistent address and mask

R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2

R5(config)#exit

R5#sho

*Aug 25 11:23:35.230: %SYS-5-CONFIG_I: Configured from console by console

R5#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 21.0.0.1

S 15.0.0.0/8 [1/0] via 21.0.0.1

S 20.0.0.0/8 [1/0] via 21.0.0.1

21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 21.0.0.0/8 is directly connected, Ethernet0/0

L 21.0.0.2/32 is directly connected, Ethernet0/0

S 30.0.0.0/8 [1/0] via 21.0.0.1

S 162.14.0.0/16 [1/0] via 192.168.168.2

S 172.19.0.0/16 [1/0] via 21.0.0.1

192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks

--More--

C:\Documents and Settings\user>tracert 162.14.0.10

Tracing route to 162.14.0.10 over a maximum of 30 hops

1	1 ms	<1 ms	<1 ms	10.0.0.1
2	1 ms	<1 ms	<1 ms	20.0.0.2
3	1 ms	<1 ms	<1 ms	30.0.0.2
4	1 ms	<1 ms	<1 ms	21.0.0.2
5	2 ms	1 ms	1 ms	192.168.168.2
6	3 ms	2 ms	1 ms	162.14.0.10

Trace complete.

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
<C> Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
<C> Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 3ms, Average = 3ms

C:\Documents and Settings\user>

Microsoft Windows XP [Version 5.1.2600]
<C> Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 162.14.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=4ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>_

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>